

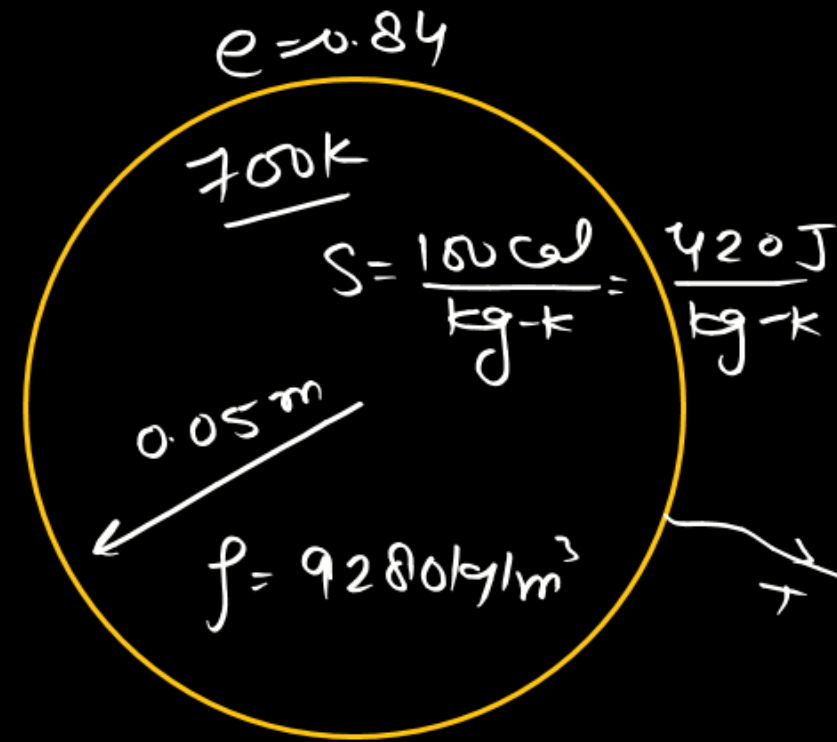
A solid sphere of diameter 0.1 m is at 427°C and is kept in an enclosure at 27°C. Take Stefan's constant

$= \frac{20}{3} \times 10^{-8} \text{ W/m}^2 \text{K}^4$, emissivity of the surface 0.84, specific heat 0.1 kcal/kg K, density = 9280 kg/m³, J = 4200 J/k

cal. If rate of decrease of temperature of the sphere is $N \times 10^{-3} \text{ }^\circ\text{C/s}$, find $\frac{N}{40}$.

$$dQ = dm L$$

$$dQ = m S dT$$



350 K

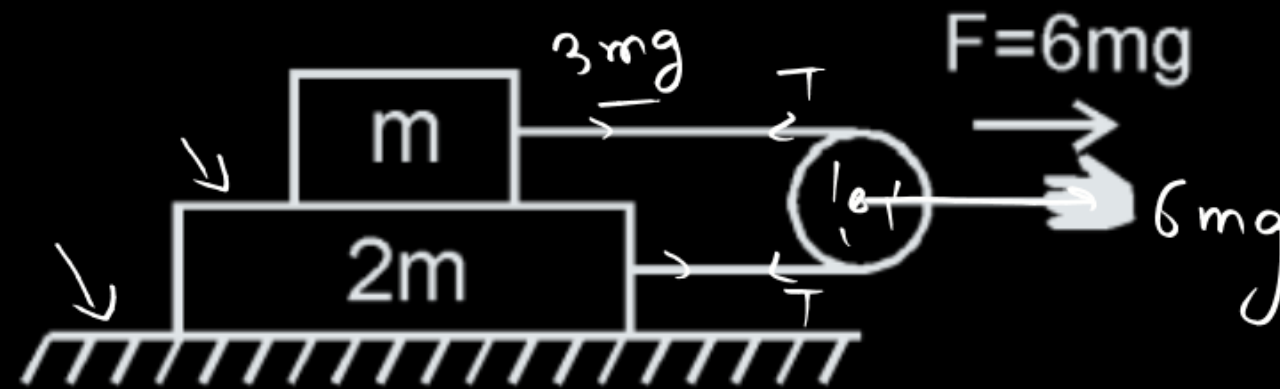
$$P = e A \sigma \{T^4 - T_s^4\} = \frac{dQ}{dt}$$

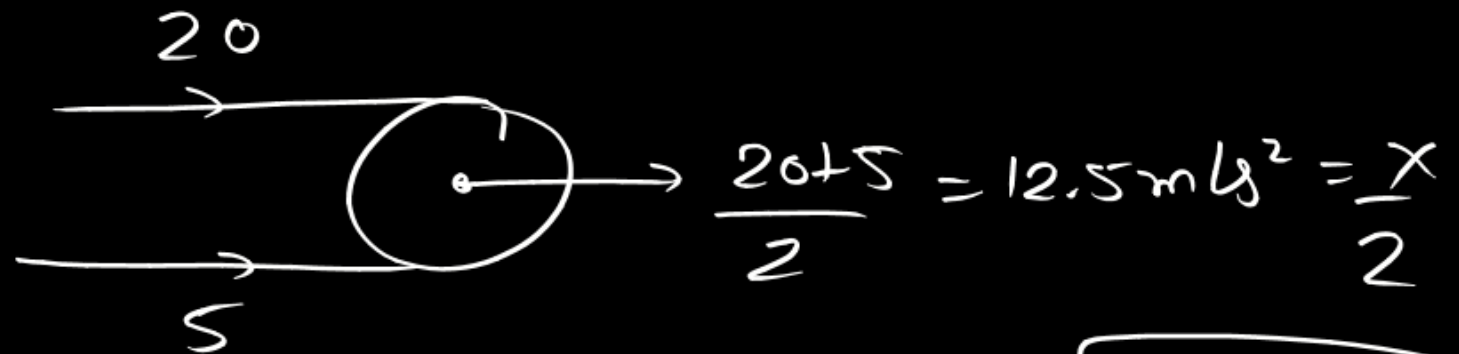
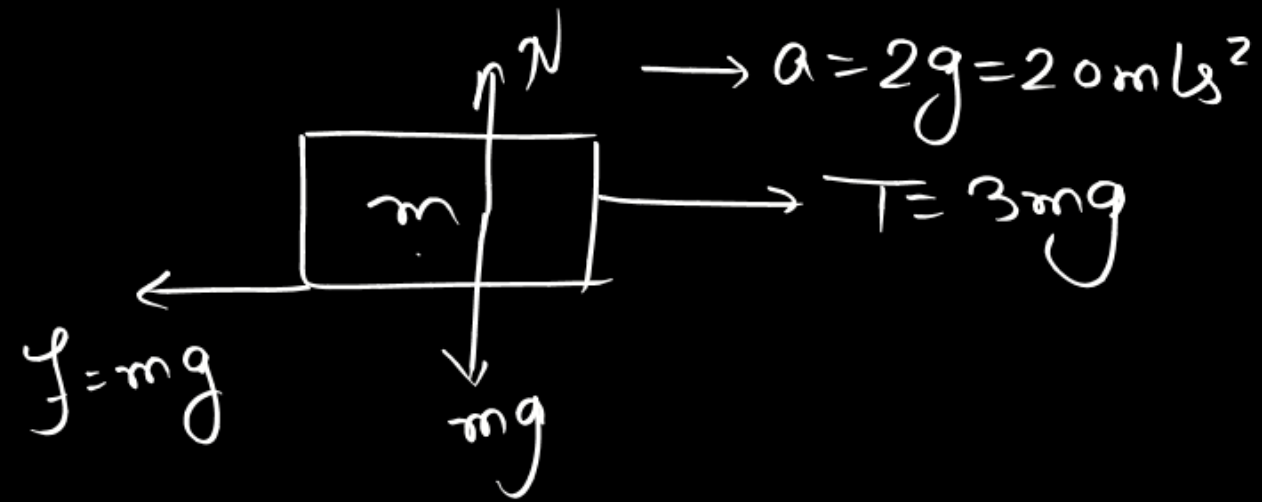
$$= e \times 4\pi r^2 \sigma (T^4 - T_s^4) = m S \left(\frac{dT}{dt} \right)$$

$$e 4\pi r^2 \sigma \{T^4 - T_s^4\} = \rho \times \frac{4}{3} \pi r^3 S \frac{dT}{dt}$$

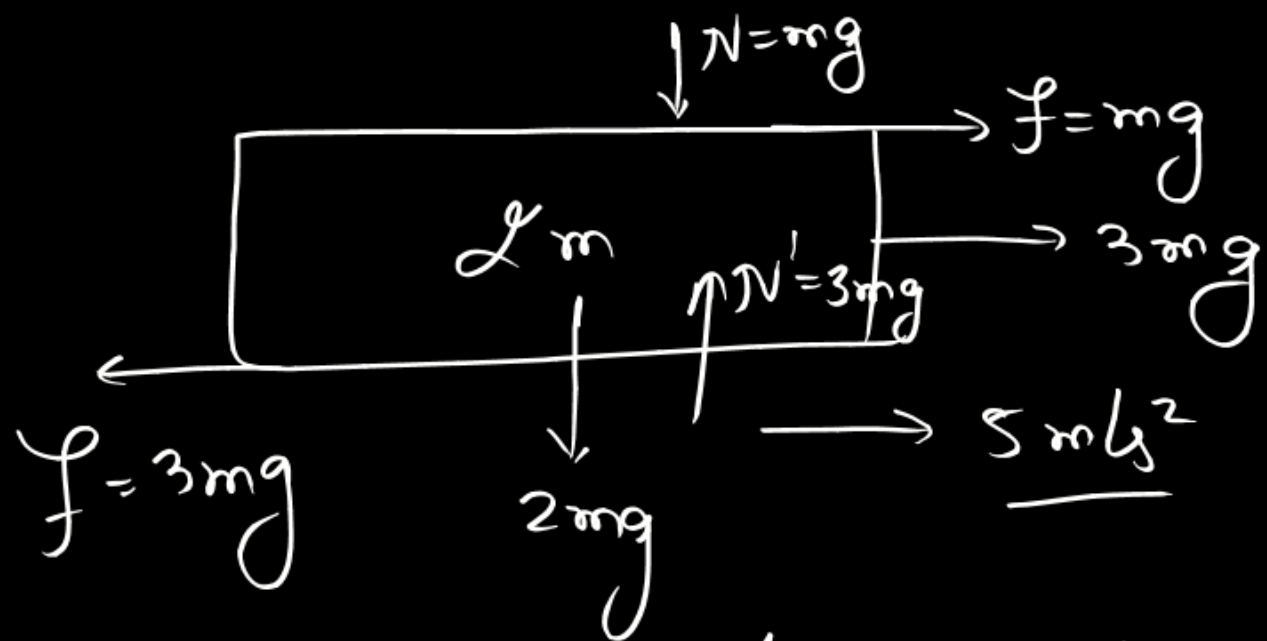
$$\frac{dT}{dt} = \frac{(200) \times 10^{-3} \text{ }^\circ\text{C}}{\text{sec}} \quad \frac{N}{40} = (5) \checkmark$$

A block of mass m is placed on top of a block of mass $2m$ which in turn is placed on fixed horizontal surface. The coefficient of friction between all surfaces is $\mu = 1$. A massless string is connected to each mass and wraps halfway around a massless and frictionless pulley, as shown. The pulley is pulled by horizontal force of magnitude $F = 6mg$ towards right as shown. If the magnitude of acceleration of pulley is $\frac{X}{2} \text{ m/s}^2$, fill the value of X . (Take $g = 10 \text{ m/s}^2$)





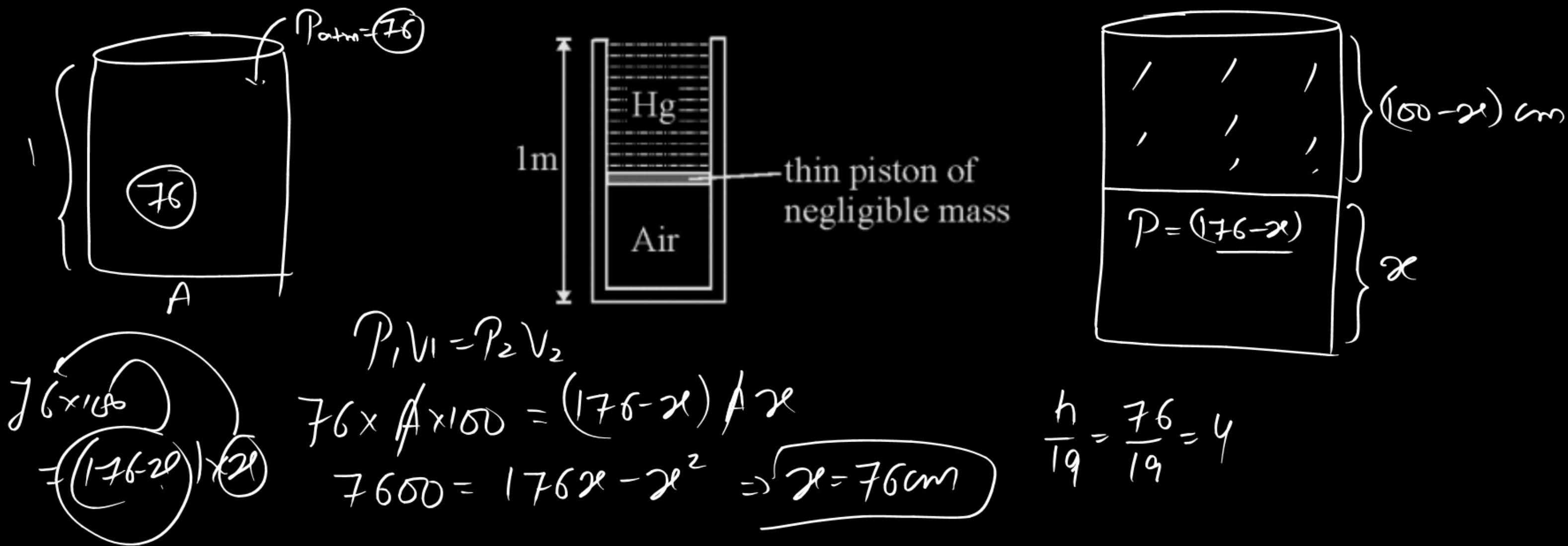
$$\boxed{X = 25}$$



$$3mg - 3mg - 2mg = (2m)a$$

$$a = g/2$$

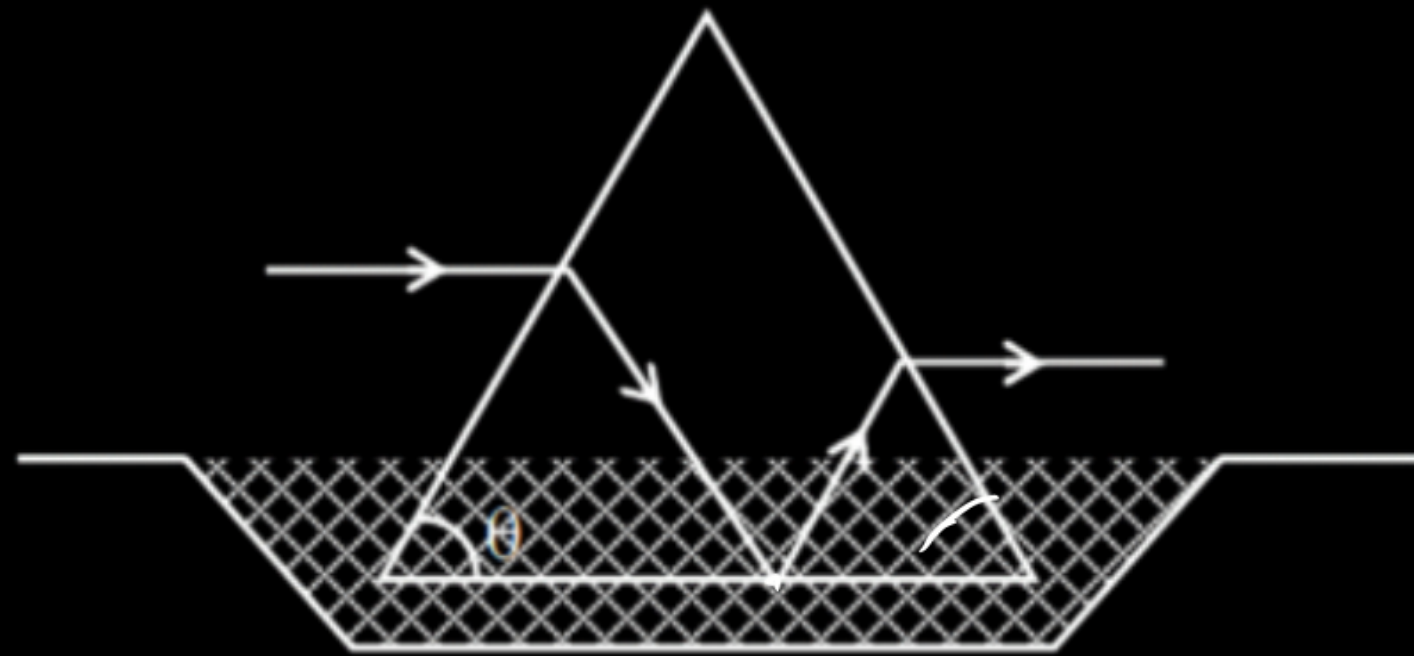
The 1 m tall conducting cylinder in figure contains air at a pressure of 76 cm of Hg. A very thin piston of negligible mass is placed at the top of the cylinder to prevent any air from escaping, then mercury is very slowly poured into the cylinder until no more can be added without the cylinder overflowing. What is the height h (in cm) of the compressed air? Fill $h/19$ in OMR



The image contains three hand-drawn diagrams and several equations:

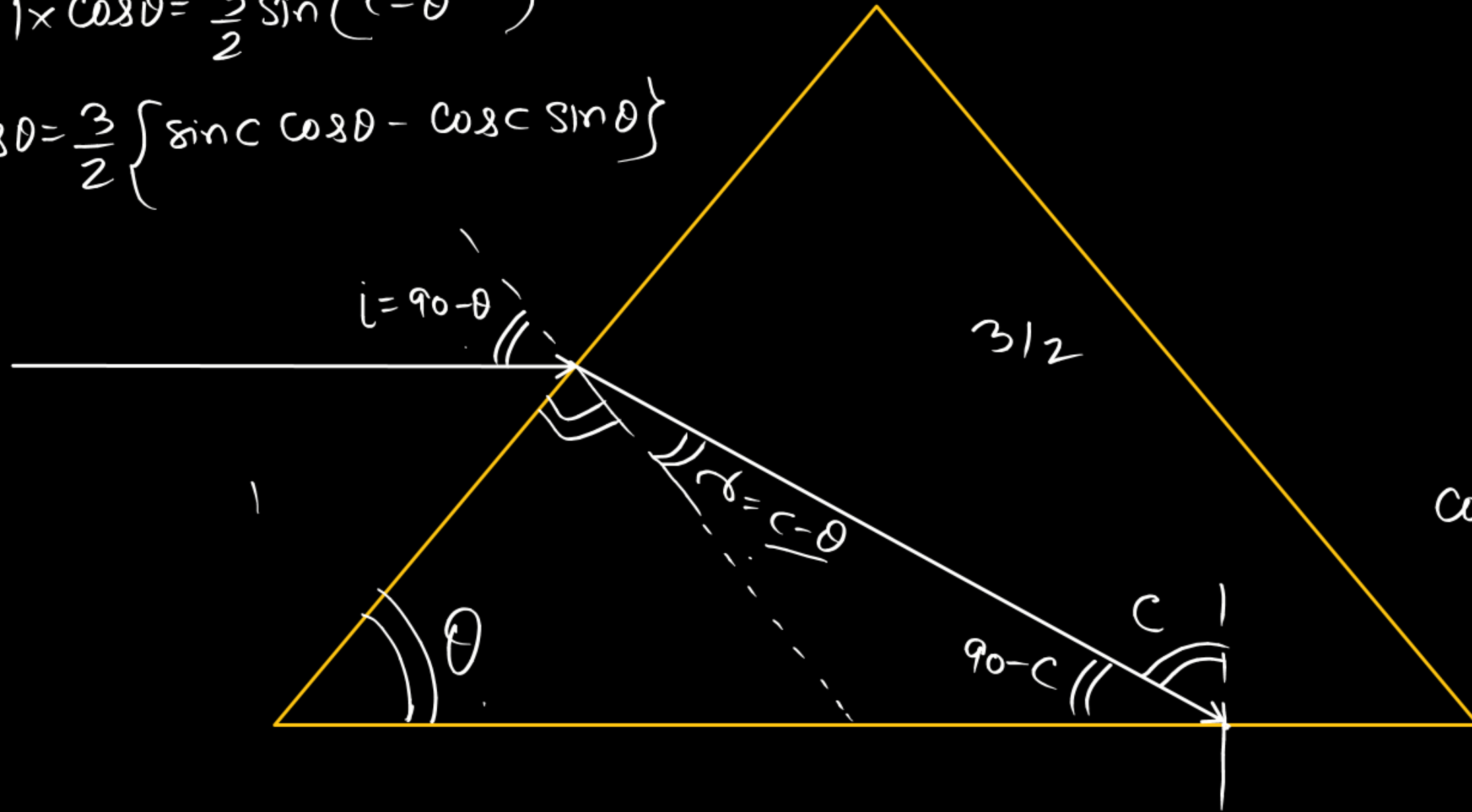
- Diagram A:** A cylinder of height 1 m. Inside, there is a circle labeled "76". Above the cylinder, an arrow points to the text $P_{atm} = 76$.
- Middle Diagram:** A cross-section of the cylinder. The bottom part is labeled "Air" and the top part is labeled "Hg". A horizontal line separates them, labeled "thin piston of negligible mass". A vertical double-headed arrow on the left indicates the total height is 1 m.
- Right Diagram:** A cylinder divided into two sections. The top section is labeled $(100 - x)$ cm and contains diagonal lines. The bottom section is labeled x and contains the text $P = (176 - x)$.
- Equations:**
 - $P_1 V_1 = P_2 V_2$
 - $76 \times A \times 100 = (176 - x) A x$
 - $7600 = 176x - x^2 \Rightarrow x = 76 \text{ cm}$ (circled)
 - $\frac{h}{19} = \frac{76}{19} = 4$
- Handwritten notes:**
 - Below Diagram A: 76×100 and $-(176 - x) \times x$ (circled).

An isosceles triangular glass prism stands with its base in water as shown. A ray is incident parallel to base of prism. After refraction through inclined face it just suffers total internal reflection on water glass interface. If value of $\tan\theta$ is $\frac{\lambda}{\sqrt{17}}$ then find λ . (Take $\mu_{\text{glass}} = 3/2$, $\mu_{\text{water}} = 4/3$)



$$1 \times \cos \theta = \frac{3}{2} \sin(c - \theta)$$

$$\cos \theta = \frac{3}{2} \{ \sin c \cos \theta - \cos c \sin \theta \}$$



$$\theta + (90 + \gamma) + 90 - c = 180$$

$$\gamma = c - \theta$$

$$\sin c = \frac{4 \times 2}{3 \times 3} = \frac{8}{9}$$

$$\cos c = \sqrt{1 - \frac{64}{81}} = \frac{\sqrt{17}}{9}$$

$$\cos \theta = \frac{3}{2} \left\{ \frac{8}{9} \times \cos \theta - \frac{\sqrt{17}}{9} \times \sin \theta \right\}$$

$$2 = \frac{8 - \sqrt{17}}{3} + \tan \theta$$

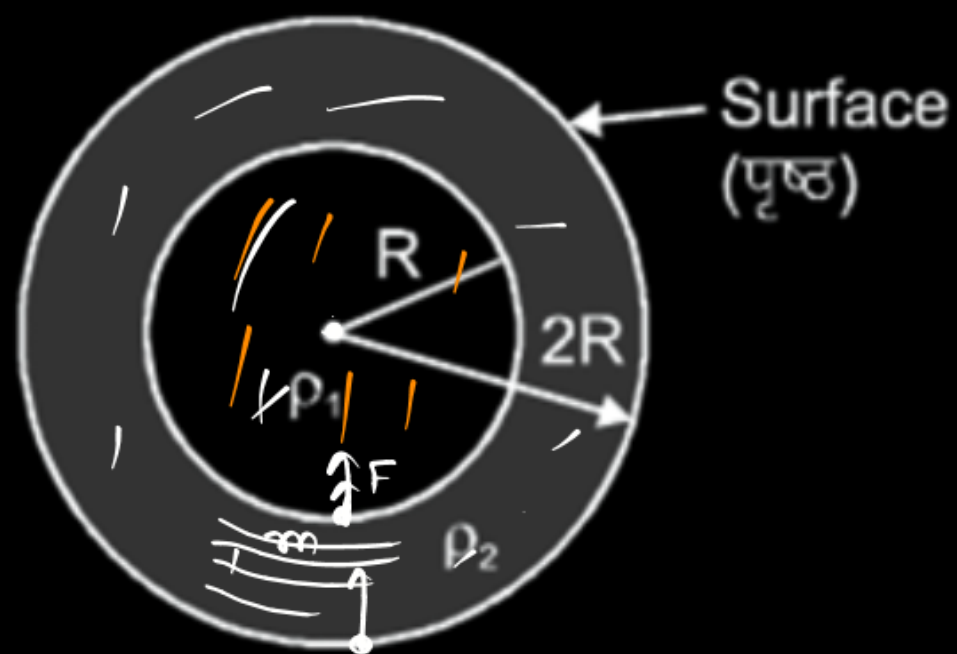
$$6 = 8 - \sqrt{17} + \tan \theta$$

$$\sqrt{17} \tan \theta = 2$$

$$\tan \theta = \frac{2}{\sqrt{17}} = \frac{\lambda}{\sqrt{17}} \Rightarrow \lambda = 2$$

A planet is made of two materials of density ' ρ_1 ' and ' ρ_2 ' as shown in figure.

$$g = \frac{F}{m} = E$$



$$g_1 = \frac{g_1 + 7g_2}{4}$$

$$3g_1 = 7g_2$$

$$\frac{\rho_1}{\rho_2} = \frac{7}{3} = \frac{x}{3} \Rightarrow x = 7$$

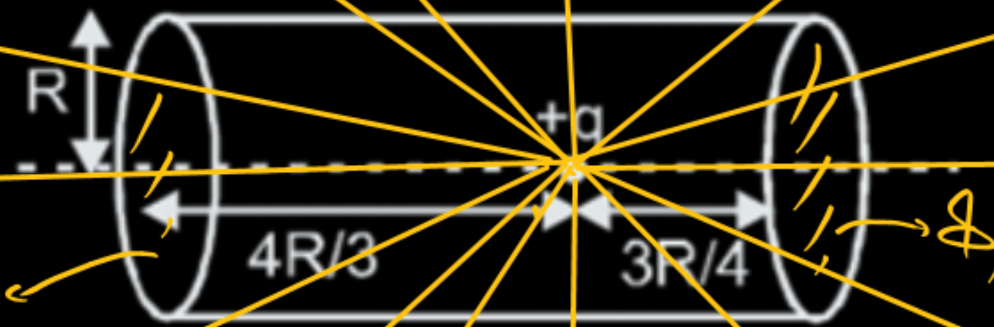
Acceleration due to gravity at surface of planet is same as that at depth ' R '. The ratio $\frac{\rho_1}{\rho_2}$ is equal to $\frac{x}{3}$.

Find the value of ' x '.

$$F = \frac{Gm}{R^2} = \frac{G \rho_1 \frac{4}{3} \pi R^3}{R^2}$$

$$F = \frac{G(m+m')}{(2R)^2} = \frac{G \left\{ \rho_1 \frac{4}{3} \pi R^3 + \rho_2 \times \frac{4}{3} \pi 7R^3 \right\}}{4R^2}$$

A point charge $+q$ is placed on the axis of a closed cylinder of radius R and length $\frac{25R}{12}$ as shown. If electric flux coming out from the curved surface of cylinder is $\frac{xq}{10\epsilon_0}$, then calculate x .



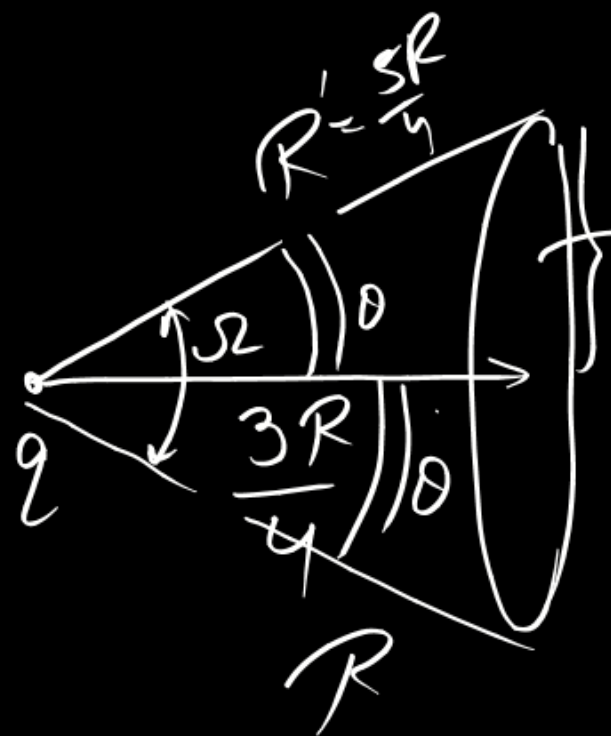
$$\Phi = \frac{q}{\epsilon_0}$$

$$\Phi_c = \Phi - \Phi_1 - \Phi_2$$

$$= \frac{q}{\epsilon_0} - \frac{q}{5\epsilon_0} - \frac{q}{10\epsilon_0}$$

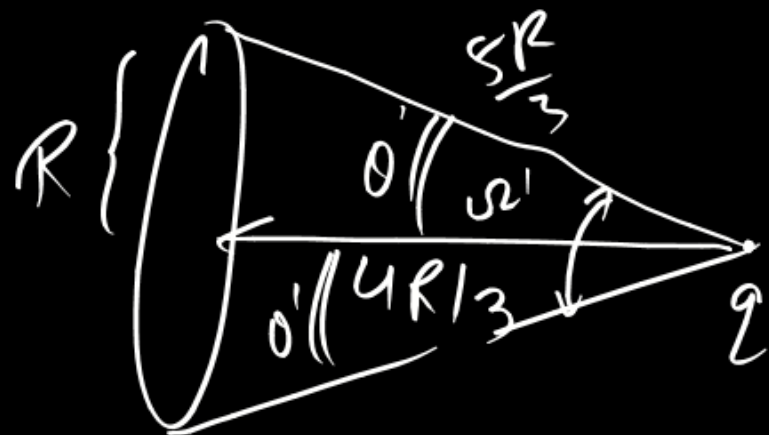
$$= \frac{10q - 2q - q}{10\epsilon_0} = \frac{7q}{10\epsilon_0}$$

$$x = 7$$



$$\sqrt{1 + \frac{9}{16}} = \frac{5R}{4}$$

$$\begin{aligned} \Phi_1 &= \frac{q}{\epsilon_0 \times 4\pi} \Omega \\ &= \frac{q}{4\pi\epsilon_0} 2\pi \left(1 - \frac{3}{5}\right) \\ &= \frac{q}{5\epsilon_0} \end{aligned}$$



$$\frac{R^2 + 16R^2}{9} = \frac{5R}{3}$$

$$\begin{aligned} &\frac{q}{\epsilon_0 \times 4\pi} \Omega' \\ &= \frac{q}{2 \times 4\pi\epsilon_0} \times 2\pi \left(1 - \frac{4R/3}{5R/3}\right) \\ &= \frac{q}{10\epsilon_0} \end{aligned}$$